ENTERPRISE TRACK A • AUTONOMOUS SMB ONBOARDING

STATUTORY BASELINE: 2026-07-15

Accel Business — Solution Architecture

Technical Architecture, 6-Layer Implementation Model, End-to-End Execution Pipeline & DPDP Compliance Framework

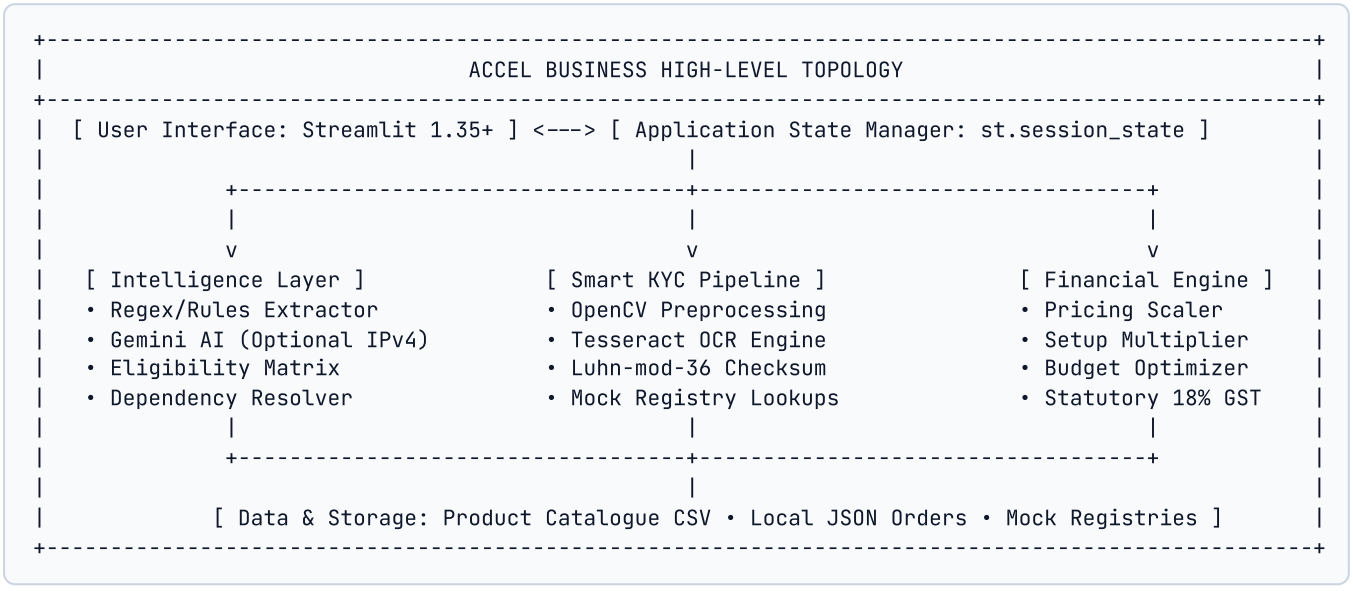
Version 1.0.0 | Architecture: Modular Monolith | 89/89 Pytest Verification Pass | Zero-Salesperson Digital Journey | DPDP Act 2023 Compliant

1. Executive Summary & Architectural Mission

The **Accel Business – Udyam Digital** platform is an autonomous onboarding engine engineered to solve India's small and medium enterprise (SMB) digital adoption barrier. Across the traditional enterprise telecommunications landscape, acquiring broadband, dedicated internet, cloud telephony, and IoT security required an average of **17.6 days**, characterized by repeated sales visits, paperwork delays, and drop-offs (**32% drop-off at needs assessment, 56% drop-off at quotation**).

Accel Business consolidates this entire sales and fulfillment cycle into a **zero-salesperson, single-sitting self-serve platform** delivering transparent pricing, verified statutory compliance, and immediate circuit provisioning.

< 10 min	100%	DPDP '23	89 / 89
SINGLE-SITTING ONBOARDING	DETERMINISTIC RULES ACCURACY	STATUTORY PRIVACY GATE	AUTOMATED QA PASS RATE



2. Complete Solution Architecture Diagram

The solution architecture is organized into **six strictly decoupled tiers**. User interactions flow from the Presentation Layer into the Application Processing Layer, dispatching to Intelligence, Financial Costing, and Local Storage, while interfacing with local and external boundary services.

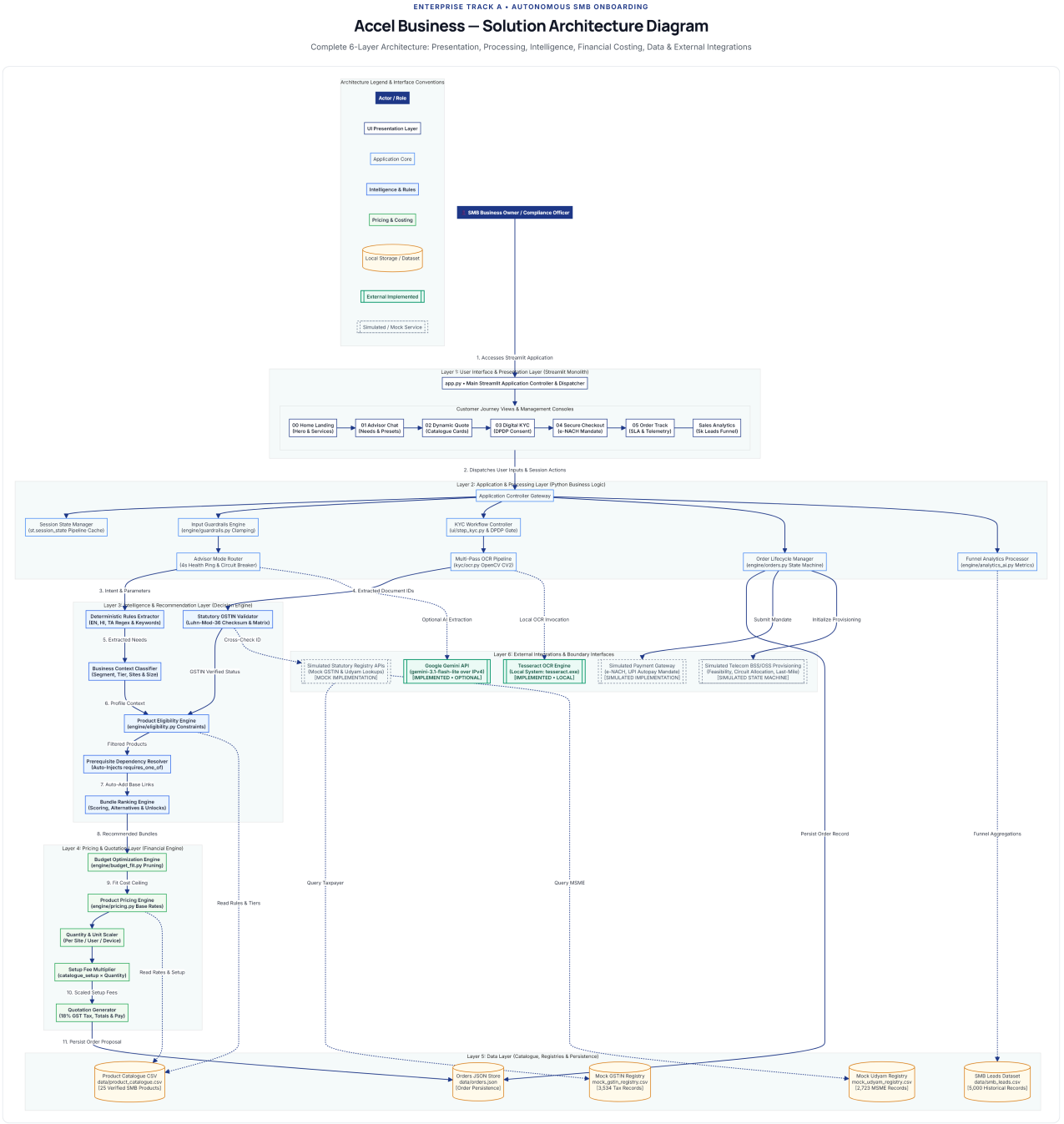


Figure 2.1: Accel Business 6-Layer Architecture Diagram (High-Resolution Top-Down Topology). Editable source available in docs/solution_architecture.mmd.

3. Six-Layer Architecture Specification

Layer 1: User Interface & Presentation Layer (Streamlit Frontend)

Built as a Python modular monolith using Streamlit 1.35+. Page state is coordinated via `app.py` and modular views in `ui/*.py`.

- **Main Application Controller (`app.py`)**: Manages top-level routing, session state initialization, and navigation switching across the 4 primary views (Home, Customer Journey, KYC Verification Desk, Sales Analytics).
- **Customer Journey Views (`ui/step_*.py`)**:
  - `00 Home Landing`: Editorial hero headline, live telemetry value strip, and feature cards.
  - `01 Advisor Chat`: Natural language requirement discovery, persona quick-fills, and digital needs slot extraction.
  - `02 Dynamic Quote`: Strictly catalogue-backed item cards, inline quantity steppers, item removal, and 18% GST calculation.
  - `03 Digital KYC`: DPDP Act 2023 statutory consent modal, multi-file uploaders, specimen selector (K01–K12), and verification audit badges.
  - `04 Secure Checkout`: Banking mandate selection (e-NACH, UPI Autopay, Corporate Net Banking), billing capture, and order authorization.
  - `05 Activation Track`: Real-time 5-stage provisioning timeline (Order Confirmed to CPE Activated) and SLA uptime monitors.
  - `Sales Analytics`: Executive BI funnel analyzing 5,000 historical leads, stage drops, and CAC metrics.
- **Editorial Design System (`ui/styles.py`)**: Deep Navy (`#1E3A8A`), Ink (`#111827`), Sand (`#E8DFCB`), Sky (`#93C5FD`), and Page Background (`#F4F5F7`). Guarantees strict ≥ 4.5:1 WCAG contrast and zero text overlap.

Layer 2: Application & Processing Layer (Python Core)

- **Session State Manager (`st.session_state`)**: Central in-memory state engine ensuring journey continuity, quote persistence, and KYC audit trails.
- **Input Guardrails Engine (`engine/guardrails.py`)**: Sanitizes inputs and clamps numeric ranges before widget rendering (employees ≤ 10,000, sites ≤ 500, budget ≤ ₹10,00,000).
- **Advisor Mode Router & Circuit Breaker (`engine/advisor.py`)**: Features a 4-second cached health check ping and a strict 10-second hard timeout on LLM calls with zero retries, automatically falling back to deterministic rules if slow.
- **KYC Verification Controller (`ui/step_kyc.py`)**: Enforces DPDP consent as a strict gatekeeper before document processing.
- **Multi-Pass OCR Pipeline (`kyc/ocr.py`)**: OpenCV image enhancement (grayscale, adaptive thresholding, morphological ROI cropping) feeding Tesseract OCR.
- **Order Lifecycle Manager (`engine/orders.py`)**: Serializes order proposals, generates unique identifiers (`AB-2026-XXXXX`), and transitions circuit states.

Layer 3: Intelligence & Recommendation Layer (Decision Engine)

Enforces 100% deterministic compliance with telecommunications catalogue rules and regulatory prerequisites:

- **Deterministic Rules Extractor (`engine/advisor.py`)**: Multilingual pattern matching across English, Hindi, and Tamil phrases, extracting employee counts, site counts, locations, and digital requirements.
- **Business Context Classifier**: Maps business profiles into telecom operational segments (`micro`, `prof`, `multi`, `mfg`) and city tiers.
- **Product Eligibility Engine (`engine/eligibility.py`)**: Evaluates constraints from `product_catalogue.csv`:
  - `min_employees` and `min_sites` thresholds.
  - `available_city_tiers`: Tier 1 vs Tier 2 eligibility. Unlisted cities default to Tier 2 with an advisory notice.
- **Statutory GSTIN Validator (`kyc/validators.py`)**: Validates 15-character GSTIN structure, computes Luhn-mod-36 checksum, and strictly blocks non-compliant products: **P03, P05, P06, P10, P12, P13, P14, P15, P21, P24**. (P19 POS and P20 Antivirus do NOT require GSTIN).
- **Prerequisite Dependency Resolver (`engine/eligibility.py`)**: Evaluates `requires_one_of` dependencies. Automatically injects required base connectivity (e.g., adding P02 Biz Fibre 300 when P11 Static IP is selected alone). Ensures P11 is never paired with leased lines (P05/P06), adhering strictly to catalogue specs.
- **Bundle Ranking & Unlock Engine (`engine/advisor.py`)**: Ranks bundles by relevance, compiles alternative add-ons, and provides actionable unlock tips.

Layer 4: Pricing & Quotation Layer (Financial Engine)

- **Product Pricing Engine (`engine/pricing.py`)**: Reads price ranges and setup fees directly from `product_catalogue.csv`.
- **Unit & Volume Scaler**: Scales units dynamically (per-site broadband, per-line postpaid, per-user cloud productivity, per-device endpoint security, per-camera CCTV).
- **Setup Fee Scaling Model**:

  > **Line Setup INR = catalogue_one_time_setup_inr × quantity**
  > Strictly applied to every product across all quantities (100% match against the 248 leads with setup fees in `smb_leads.csv`).

- **Budget Optimization Engine (`engine/budget_fit.py`)**: Iteratively prunes optional items to fit the customer's budget ceiling while strictly preserving base connectivity and prerequisite dependencies.
- **Quotation Generator (`engine/pricing.py`)**: Computes subtotal recurring monthly fees, subtotal one-time setup fees, statutory 18% GST:

  > **Statutory Tax = (Monthly Subtotal + One-Time Setup) × 0.18**

  Computes First Month Payable total, contract duration, and estimated installation lead times.

Layer 5: Data Layer (Catalogue, Registries & Persistence)

- **Product Catalogue (`data/product_catalogue.csv`)**: Ground truth defining 25 Accel Business products, pricing bounds, contract terms, tiers, and dependency rules.
- **Mock GSTIN Registry (`data/mock_gstin_registry.csv`)**: 3,534 synthetic tax records for API lookups.
- **Mock Udyam Registry (`data/mock_udyam_registry.csv`)**: 2,723 synthetic MSME registration records.

- **Historical Leads Dataset (`data/smb_leads.csv`)**: 5,000 synthetic leads powering conversion funnel analysis and CAC benchmarks.
- **Orders JSON Store (`data/orders.json`)**: Immutable JSON datastore persisting placed orders and tracking state.

Layer 6: External Integrations & Boundary Interfaces

Service / Integration	Category	Implementation Status	Technical Mechanism
Google Gemini API	Generative AI	**Implemented (Optional)**	`gemini-3.1-flash-lite` over IPv4 (`0.0.0.0`) with 10s strict timeout and zero retries.
Tesseract OCR	Computer Vision	**Implemented (Local System)**	Local binary (`tesseract.exe`) via `pytesseract` with OpenCV thresholding & ROI cropping.
Statutory Registries	Govt Verification	**Mock Service**	In-memory lookup against mock registries with Luhn-mod-36 checksum verification.
Banking Gateway	Payment Mandate	**Simulated Service**	UI-driven mandate setup simulating e-NACH, UPI Autopay, and Net Banking authorization.
Telecom BSS/OSS	Provisioning	**Simulated State Machine**	State machine tracking 5 provisioning stages with live circuit telemetry updates.

4. End-to-End Execution Pipeline (14 Stages)

> **Chronological Execution Order:** User Entry → Requirement Input → Guardrail Clamping → Hybrid AI/Rules Routing → Context Analysis → Eligibility Validation → Dependency Injection → Bundle Ranking → Budget Optimization → Setup Fee Scaling → 18% GST Quote Generation → DPDP Consent & OCR → Mandate Authorization → Circuit Provisioning Telemetry.

1. **Application Access**: SMB owner launches the Streamlit application; `init_session()` establishes persistent session state.
2. **Requirement Submission**: User enters operational parameters (employees, sites, location, GSTIN) and natural language requirements in English, Hindi, or Tamil.
3. **Input Guardrail Clamping**: `engine/guardrails.py` clamps values to valid ranges (1 to 10,000 employees, 1 to 500 sites, ₹0 to ₹10,00,000 budget).
4. **Hybrid Extraction Routing**:
   - If *AI mode (Gemini)* is ON, a cached 4s health check runs; queries `gemini-3.1-flash-lite` over IPv4 with a 10s hard timeout.
   - If AI mode is OFF or times out, the deterministic rules engine extracts parameters instantly with zero latency.
5. **Context & Segment Classification**: Evaluates operational scale and assigns customer segment (`micro` , `prof` , `multi` , `mfg`).
6. **Product Eligibility Filtering**: Validates catalogue rules against size and tier. Unlisted cities default to Tier 2 with an advisory badge.
7. **Prerequisite Dependency Injection**: Auto-injects prerequisite base connectivity (e.g. P02 Biz Fibre 300) when dependent products (P11 Static IP, P12 SIP) are selected alone.
8. **Bundle Ranking & Unlock Synthesis**: Ranks candidate products by relevance score and compiles upgrade tips.
9. **Budget Ceiling Optimization**: Prunes optional items if the monthly total exceeds the budget ceiling while strictly preserving base links.
10. **Pricing & Setup Fee Scaling**: Calculates recurring costs and scales setup charges as `catalogue_one_time_setup_inr × quantity` across all products.
11. **Quotation Generation & Review**: Generates `QuoteResult` with statutory 18% GST, contract commitments, and installation lead times.
12. **DPDP Consent Gate**: The user must explicitly accept the statutory Digital Personal Data Protection notice before proceeding.
13. **Multi-Pass OCR & Compliance Validation**: Documents are preprocessed via OpenCV and parsed via Tesseract OCR; GSTINs and Udyam numbers are checksum-validated and cross-referenced.
14. **Checkout Mandate & Circuit Provisioning**: User authorizes an e-NACH/UPI mandate; an immutable order record (`AB-2026-XXXXX`) is persisted to `data/orders.json` and real-time circuit provisioning telemetry is initialized.

5. Security, Privacy & DPDP Compliance Architecture

- **DPDP Act 2023 Mandatory Consent Gate**: Prior to document submission, users must actively confirm statutory consent. Unconsented files are rejected at the UI boundary.
- **PII Masking in UI and Logs**: Aadhaar numbers are masked to the last 4 digits (`XXXX-XXXX-1234`); PAN numbers are redacted from public audit logs.
- **Local Document Processing**: All OCR and validation operations run locally; documents are never transmitted to third-party cloud extraction providers.
- **Zero Usage Telemetry**: Streamlit telemetry is explicitly disabled in `.streamlit/config.toml` (`gatherUsageStats = false`).
- **Repository Secret Hygiene**: `.env` and customer datasets are strictly excluded via `.gitignore` . Only `.env.example` is committed.

6. Technical Tech Stack Matrix

Layer / Component	Technology	Version	Purpose
Language Runtime	Python	3.15.0+	Core application execution runtime
Web Framework	Streamlit	1.35.0+	Modular monolith UI, reactive state & steppers
Data Processing	Pandas	2.2.0+	Catalogue parsing, registry lookups & analytics
Generative AI	Google GenAI SDK	0.1.1+	Multilingual extraction (`gemini-3.1-flash-lite`)
Computer Vision	OpenCV Headless	4.9.0+	Image enhancement, adaptive thresholding & ROI crops
OCR Engine	Pytesseract / Tesseract	0.3.10+ / 5.x	Local binary OCR character recognition
Data Visualization	Plotly	5.20.0+	Interactive conversion funnels & leakage charts
Test Framework	Pytest	8.0.0+	89-test verification suite (eligibility, pricing, fuzzing)
Browser Automation	Playwright	1.40.0+	Layout overlap verification & PDF rendering
Diagramming Standard	Mermaid.js	10.x	Solution architecture topological modeling

7. Quality Assurance & Verification Results

- **Pytest Automated Test Suite**: **89 / 89 tests passing (100%)** in `pytest -v` , covering multilingual extraction, boundary fuzzing (10,000 employees, 500 sites), catalogue dependency validation, setup fee scaling across 248 leads, and WCAG AAA/AA visual contrast ratios (≥ 4.5:1).
- **Playwright Layout Verification (`scripts/verify_layout_overlap.py`)**: Clean exit code 0 across Desktop (1440px), Landscape Tablet (1024px), Portrait Tablet (768px), and Mobile (390px) viewports with **zero bounding box overlaps, zero clipped text, and zero horizontal page overflow**.